5.4 Nuclear Decay

This topic covers radioactive decay.

This topic may be studied using applications that relate to, for example, medical physics and carbon dating.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

133	understand the concept of *nuclear binding* energy and be able to use the equation $\Delta E = c^2 \Delta m$ in calculations of nuclear mass (including mass deficit) and energy
134	use the *atomic mass unit (u)* to express small masses and convert between this and SI units
135	understand the processes of nuclear fusion and fission with reference to the binding energy per nucleon curve
136	understand the mechanism of nuclear fusion and the need for very high densities of matter and very high temperatures to bring about and maintain nuclear fusion
137	understand that there is background radiation and how to take appropriate account of it in calculations
138	understand the relationships between the nature, penetration, ionising ability and range in different materials of nuclear radiations (alpha, beta and gamma)
139	be able to write and interpret nuclear equations given the relevant particle symbols
140	**CORE PRACTICAL 15: Investigate the absorption of gamma radiation by lead**
141	understand the spontaneous and random nature of nuclear decay
142	be able to determine the half-lives of radioactive isotopes graphically and be able to use the equations for radioactive decay activity $A = \lambda N$, $\frac{dN}{dt} = -\lambda N$, $\lambda = \frac{\ln 2}{t_{1/2}}$, $N = N_0 e^{-\lambda t}$ and $A = A_0 e^{-\lambda t}$ and derive and use the corresponding log equations.